SAFE-CARE™ Healthcare AI Agent Safety Case Template

A structured, evidence-backed argument that an agent is safe enough for its intended use

SAFE-CARE Framework · authored by Yassen Eltayeb, Conefia · Version 1.2.0 · July 2026

Practical example: When a hospital buyer asks, “How do you know this agent will not give unsafe advice?”, the safety case answers, because it connects claims, risks, controls, evidence, and open issues.

Safety case structure

A safety case has ten sections, from intended use to signoff.

Section	What to write
1. Intended use	Who uses the agent, for what tasks, in what setting, with what limitations?
2. Safety claims	What must be true for the agent to be considered safe enough?
3. Hazard analysis	What can go wrong? Use the risk taxonomy.
4. Controls	Which guardrails, tool constraints, memory policies, and escalation rules mitigate each risk?
5. Evidence	Which tests, simulations, judge scores, clinician reviews, and production metrics support each claim?
6. Human oversight	When and how humans review, intervene, and approve releases.
7. Residual risk	What risks remain and why they are acceptable

Section	What to write
	for intended use.
8. Monitoring plan	How production signals detect drift, incidents, and regressions.
9. Incident response	How severe failures are triaged, fixed, reported, and converted into regression tests.
10. Signoff	Product, AI, clinical, privacy/legal, and operations approval.

Safety claim examples

Each claim below names the evidence it requires.

Claim	Evidence required
The agent does not provide prescription-like guidance.	Medication scenario suite, output guardrail logs, clinician review of failures.
The agent escalates red flags reliably.	100% pass on critical scenarios, read in v1.3.0 as zero misses across all K repeat runs (`docs/07`); production escalation monitoring.
The agent grounds medical claims in approved evidence.	RAG trace audit, citation judge score, clinician spot-check.
The agent writes sensitive health profile fields only after confirmation.	Tool-call logs, schema validation, confirmation UI screenshots.
The agent uses memory without stale or invented facts.	Time-series tests, contradiction scenarios, user correction logs.

Risk-control-evidence matrix

The matrix ties each risk to a control, evidence and owner.

Risk	Control	Evidence	Owner
Missed red flag	Input classifier + output escalation check	Critical scenario suite pass rate	AI Safety
Unsafe medication advice	No-prescription prompt + output guardrail	HRT/medication scenario review	Clinical + AI
False citation	Citation support judge	Citation QA report	AI Evaluation
Bad profile write	Confirmation + schema validation	Tool trace audit	Backend + AI
Privacy leakage	Data minimization + retention policy	Privacy review + log audit	Privacy/Legal

Residual risk acceptance

Remaining risk is accepted in writing, with an owner and expiry.

Residual risk	Why it remains	Compensating control	Owner	Acceptance / expiry
Example: occasional false-positive escalation	Sensitivity prioritized for a high-risk symptom class	Human review; user-friendly wording; monitor escalation rate	Clinical safety lead	Accepted for pilot; review after 30 days
[Risk]	[Rationale]	[Control]	[Owner]	[Decision/date]

Safety claim pattern

Based on these tables, write every claim as: “For [intended use and population], configuration [version] is acceptably safe with respect to [specific hazard] because [controls] achieved [measured evidence], subject to [limitations and monitoring].” In practice, each slot forces a named hazard, a configuration and measured evidence, so the pattern rules out vague claims such as “the agent is safe.”

Required review questions

Before signoff, test every claim against these five questions.

- What evidence would falsify this safety claim?
- Which user group or context is not represented in the evidence?
- What happens when retrieval, tools, or escalation services are unavailable?
- Who can stop the system, and how quickly?
- When does the safety case expire or require re-approval?

Companion documents

Also published in the SAFE-CARE public release repository (github.com/Conefia/SAFE-CARE · DOI 10.5281/zenodo.21330841):

- **Menopause-support companion design case study**: the first applied case study of SAFE-CARE (docs/03_SAFE-CARE_Menopause-Support_Companion_Case_Study).
- **Building Safe Conversational AI Agents for Healthcare**: conference talk outline (docs/09_SAFE-CARE_Conference_Talk_Deck_Outline).

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